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$$f(x) = \ln \left(\sqrt{\frac{1 + \cos(ax)}{1 - \cos(ax)}} \right)$$

$$f(x) = \ln \left(\left(\frac{1 + \cos(ax)}{1 - \cos(ax)} \right)^{\frac{1}{2}} \right)$$

$$f(x) = \frac{1}{2} \ln \left(\frac{1 + \cos(ax)}{1 - \cos(ax)} \right)$$

$$f(x) = \frac{1}{2} (\ln(1 + \cos(ax)) - \ln(1 - \cos(ax)))$$

$$f'(x) = \frac{1}{2} \left(\frac{1}{1 + \cos(ax)} \cdot (-a \sin(ax)) - \frac{1}{1 - \cos(ax)} \cdot (a \sin(ax)) \right)$$

$$f'(x) = \frac{-a \sin(ax)}{2} \left(\frac{1}{1 + \cos(ax)} + \frac{1}{1 - \cos(ax)} \right)$$

$$f'(x) = \frac{-a \sin(ax)}{2} \left(\frac{(1 - \cos(ax)) + (1 + \cos(ax))}{(1 + \cos(ax))(1 - \cos(ax))} \right)$$

$$f'(x) = \frac{-a \sin(ax)}{2} \left(\frac{2}{1 - \cos^2(ax)} \right)$$

$$f'(x) = \frac{-a \sin(ax)}{2} \cdot \frac{2}{\sin^2(ax)}$$

$$f'(x) = \frac{-a}{\sin(ax)}$$

References